

Question	Answer	Marks
1(a)(i)	<i>either</i> direction of force on a (small test) mass <i>or</i> direction of acceleration of a (small test) mass	B1
1(a)(ii)	Any three from: - the lines are radial - near the surface the lines are (approximately) parallel - parallel lines so constant field strength - constant field strength hence constant acceleration of free fall	B3
1(b)(i)	$g = GM/R^2$ $g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^3 \times 10^3)^2$	C1
	$g = 1.62 \text{ N kg}^{-1}$	A1
1(b)(ii)	<i>either</i> $x\omega^2 = GM/x^2$ <u>and</u> $\omega = 2\pi / T$ <i>or</i> $v^2/x = GM/x^2$ <u>and</u> $v = 2\pi r / T$	C1
	$(1.74 \times 10^6 + 320 \times 10^3)^3 \times 4\pi^2 / T^2 = (6.67 \times 10^{-11} \times 7.35 \times 10^{22})$	C1
	$T^2 = 7.04 \times 10^7$ $T = 8400 \text{ s (8390)}$	A1

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2(c)	pH = 8.7	01